

7 FALAR DUAS LÍNGUAS TORNA- NOS MAIS INTELIGENTES?

Psicologia no Quotidiano
Opção UMinho

MOMENTO PARA RESPONDER E REFLETIR

(MR²)

Falar duas línguas torna-nos mais
inteligentes?

CASOS PRODIGIOSOS DE PESSOAS QUE FALAM VÁRIAS LÍNGUAS



<https://www.youtube.com/watch?v=Km9-DiFaxpU>

ENTRAR

TWELVE COMMON MYTHS & MISCONCEPTIONS

ABOUT BILINGUAL CHILDREN

BY BARBARA ZURER PEARSON

Reprinted with permission from *Raising a Bilingual Child*, by Barbara Zurer Pearson, PhD., p.300

These are 12 myths & misconceptions...

1. Bilingual children start to speak later than monolinguals.

myth

3. Young children soak up languages like sponges.

myth

2. Bilinguals start out school behind monolinguals and they never catch up.

myth

6. If bilinguals score lower on standardized tests, it shows they have lower aptitude than the average monolingual child.

4. Bilinguals are just like two monolinguals in one person.

myth

5. You have to be gifted in languages in order to learn two languages at once.

myth

myth

7. Latino immigrants in the U.S. resist learning English and want everyone to learn to speak Spanish.

myth

8. Some languages are more primitive than others and are therefore easier to learn. The reason so many people can speak English is that English has less grammar than other languages.

myth

9. Speaking a second language is its own reward.

myth

11. If a bilingual child experiences any language problems in one or both languages, dropping one of the languages will fix the situation.

myth

10. Parents who do not speak a language perfectly will pass their errors and their accent on to their children.

myth

12. There's only one right way to raise a bilingual child.

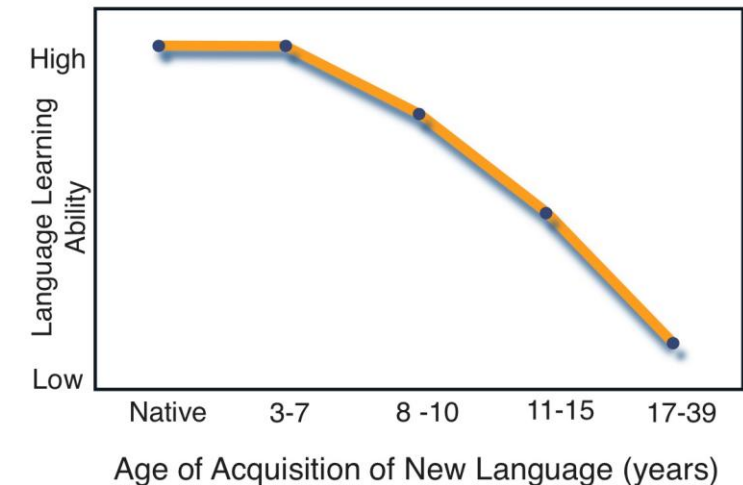
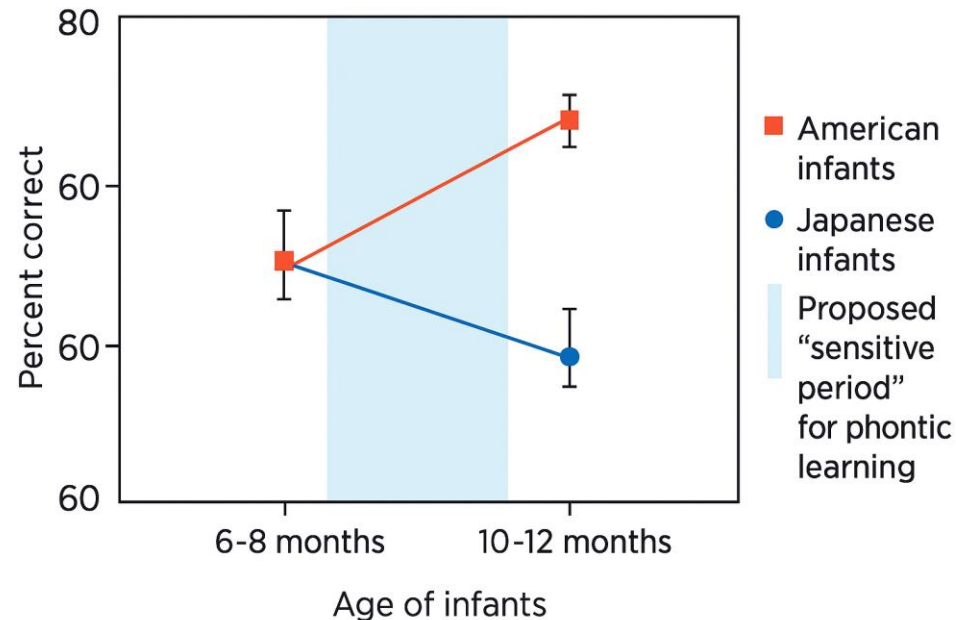
myth

FALAR DUAS LÍNGUAS TORNA-NOS MAIS INTELIGENTES?

- A linguagem é um dos melhores exemplos da existência de períodos “críticos” (ou sensíveis) no desenvolvimento
 - i.e., um período durante o qual os bebês estão mais sensíveis às características das línguas que ouvem à sua volta e conseguem “ajustar” o seu cérebro para responder de forma equivalente a ambas as línguas (ver Kuhl et al., 2008; Kuhl & Ferjan-Ramírez, 2019; Werker & Hensch, 2015).



Percepção de contrastes fonéticos: Head turn Preference Paradigm (HPP)



Creating bilingual minds

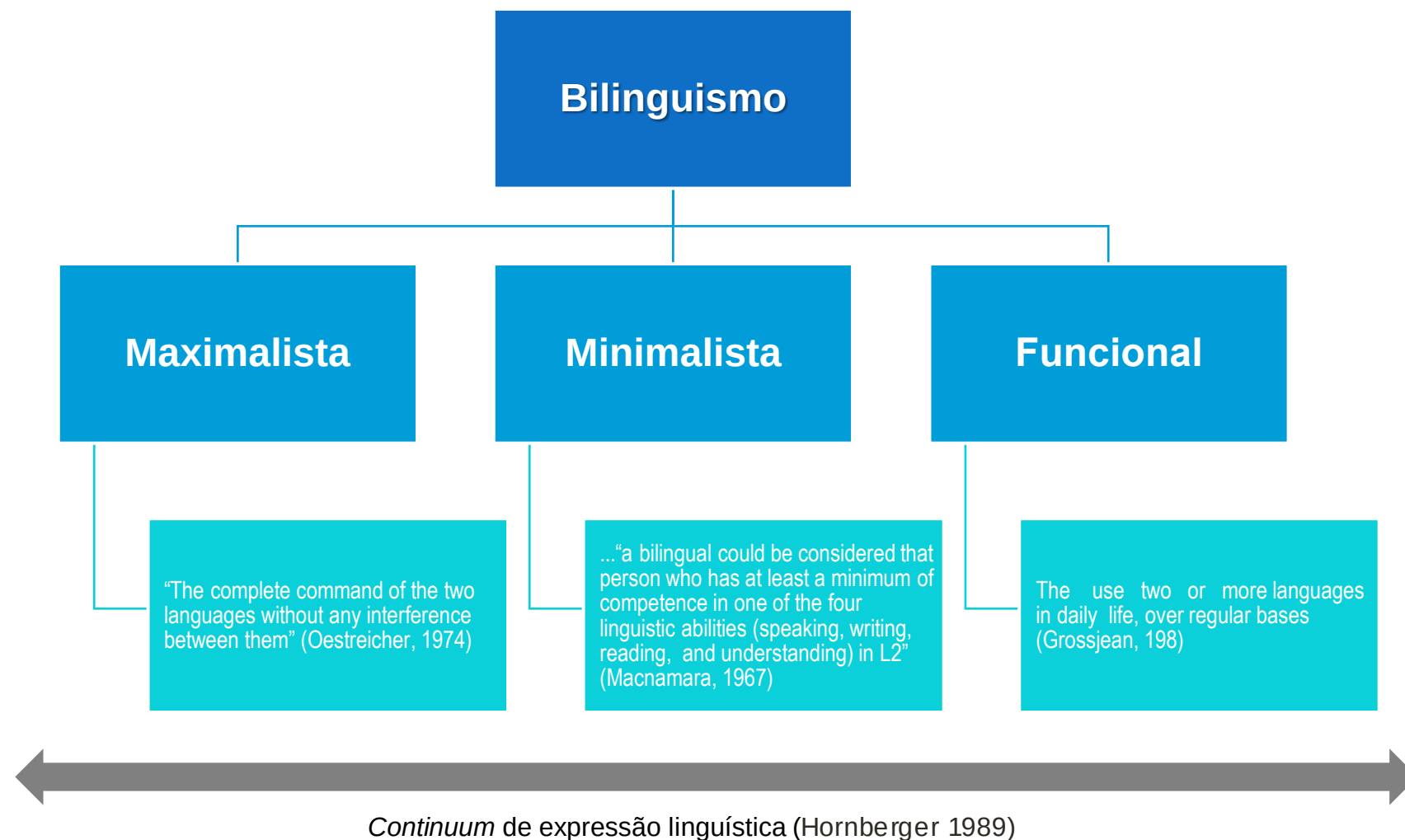
Naja Ferjan Ramirez



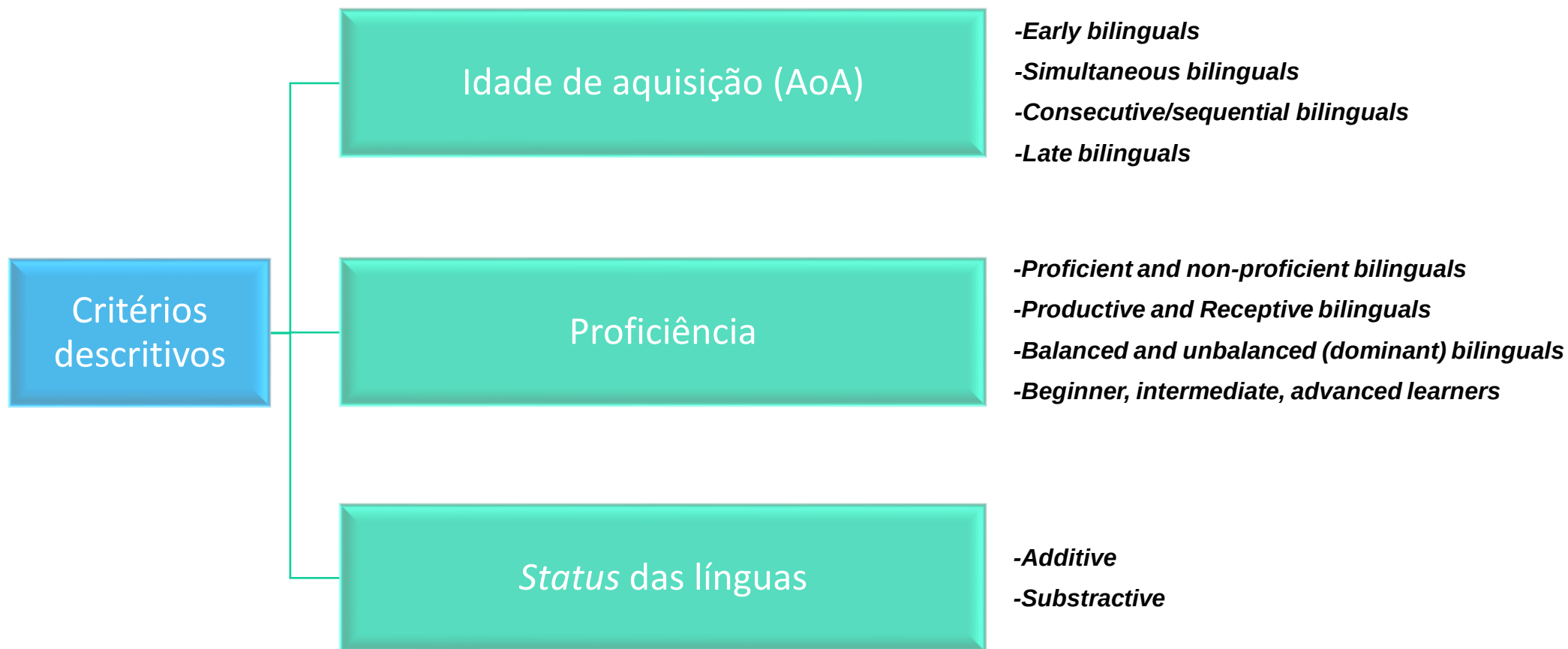
<https://www.youtube.com/watch?v=Bp2Fvkt-TRM>

O QUE SIGNIFICA SER BILINGUE?

- Questão complexa...



O QUE SIGNIFICA SER BILINGUE?



BILINGUES: IDADE DE AQUISIÇÃO

Bilinguismo simultâneo

L1 e L2 aprendidas mais ou menos ao mesmo tempo na infância.

Bilinguismo sequencial precoce

L1 aprendida primeiro, mas L2 aprendida relativamente cedo.

O maior grupo do mundo.

Bilinguismo tardio

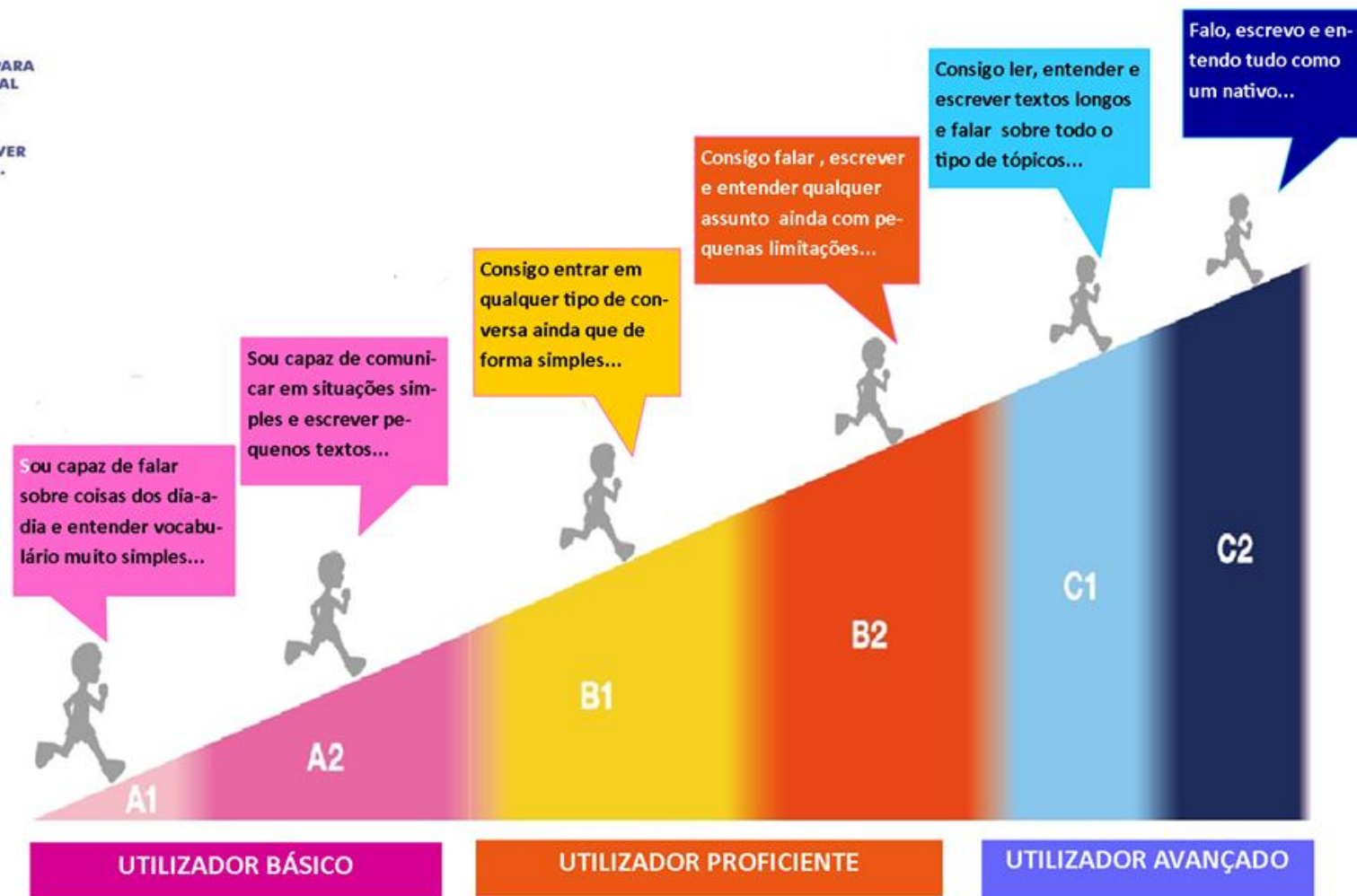
L1 aprendida primeiro e L2 aprendida após adolescência.

BILINGUES: PROFICIÊNCIA



O QUADRO EUROPEU COMUM DE REFERÊNCIA PARA LÍNGUAS (CEFR) É UM PADRÃO INTERNACIONAL UTILIZADO PARA DESCREVER HABILIDADES LINGÜÍSTICAS.

É UTILIZADO NO MUNDO TODO PARA DESCREVER AS HABILIDADES DE UM ALUNO NA LÍNGUA.



BILINGUES: PROFICIÊNCIA



O QUADRO EUROPEU COMUM DE REFERÊNCIA PARA LÍNGUAS (CEFR) É UM PADRÃO INTERNACIONAL UTILIZADO PARA DESCREVER HABILIDADES LINGÜÍSTICAS.

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Principiante

- baixa proficiência na L2 porque os bilingues estão nas primeiras fases de aquisição da L2 (e.g., níveis A1, A2)

Intermédio

- proficiência média na L2 porque os bilingues estão em fases intermédias de aquisição da L2 (e.g., níveis B1, B2)

Avançado

- elevada proficiência na L2 porque os bilingues estão em fases avançadas da aquisição L2 (e.g., níveis C1, C2)

TERMINOLOGIA

Desequilibrado:
bilingues com grande domínio nas duas línguas, mas que usam significativamente mais uma do que a outra (língua dominante).

Proficiente:
bilingues com grande domínio nas duas línguas, mas não necessariamente equilibrado.

Não-proficientes:
bilingues com maior domínio numa língua do que na outra.

Equilibrado:
bilingues com domínio equivalente das duas línguas.

Produtivos:
compreendem e produzem nas duas línguas.

Recetivos:
compreendem nas duas línguas mas têm capacidades de produção mais limitadas.

BILINGUES: STATUS DAS LÍNGUAS



Aditivo

- a aprendizagem da L2 não interfere com a aprendizagem da L1.
- as duas línguas estão bem desenvolvidas e são socialmente valorizadas.

Subtrativo

- a aprendizagem da L2 interfere com a aprendizagem da L1.
- uma das línguas é desvalorizada (pressão social para não a utilizar).
- a L2 substitui a L1.





The Language History Questionnaire (LHQ)

is an important tool for assessing the language background of learners/speakers of multiple languages.

[SIGN UP NOW](#)

LHQ3 works the best when you use the following features



MULTIPLE LANGUAGES

Thanks to [native speaker](#)



AGGREGATED SCORES

Four aggregated scores can



VISUALIZATION

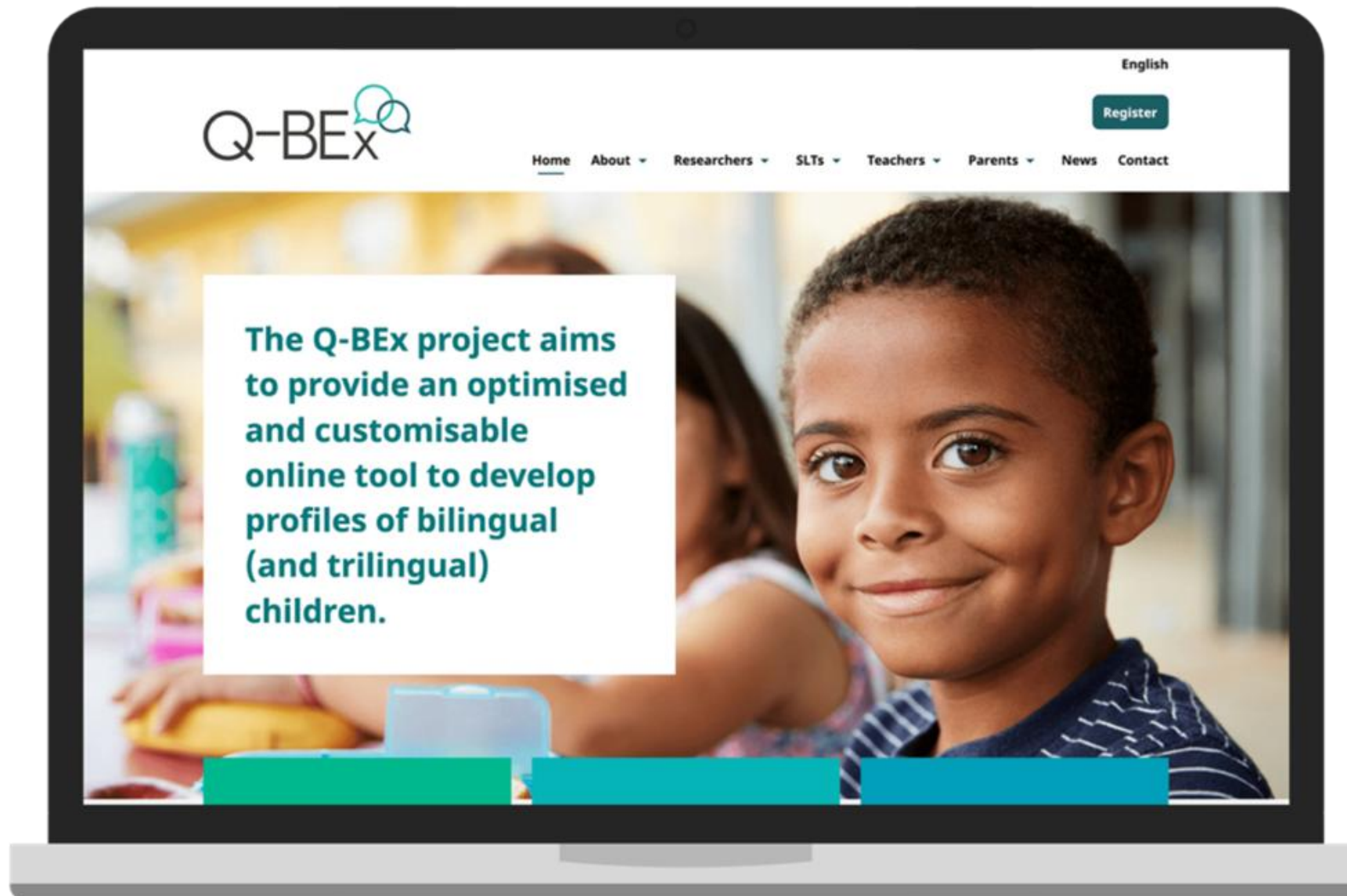
LHQ3 automatically

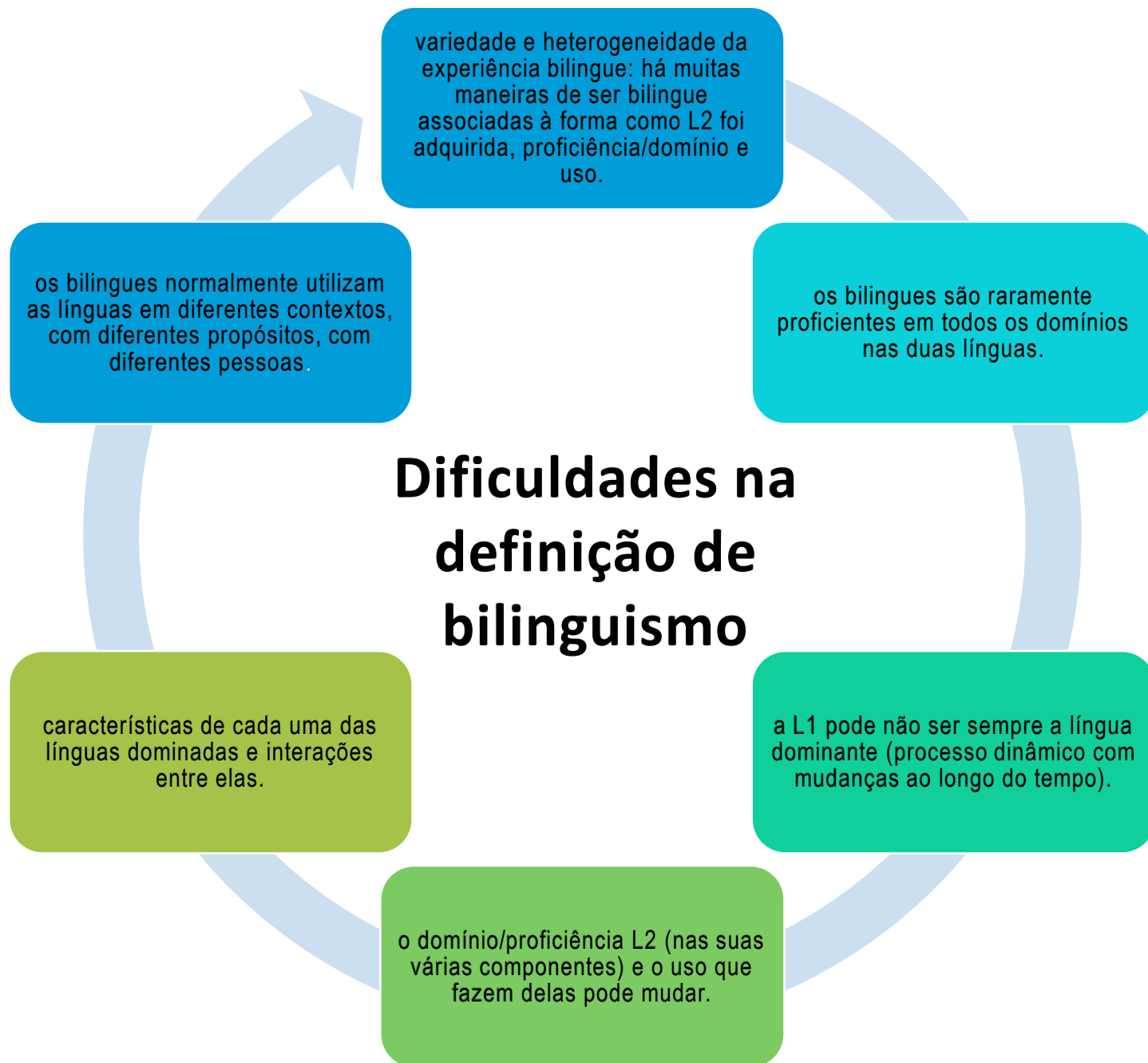


MULTI-PLATFORM

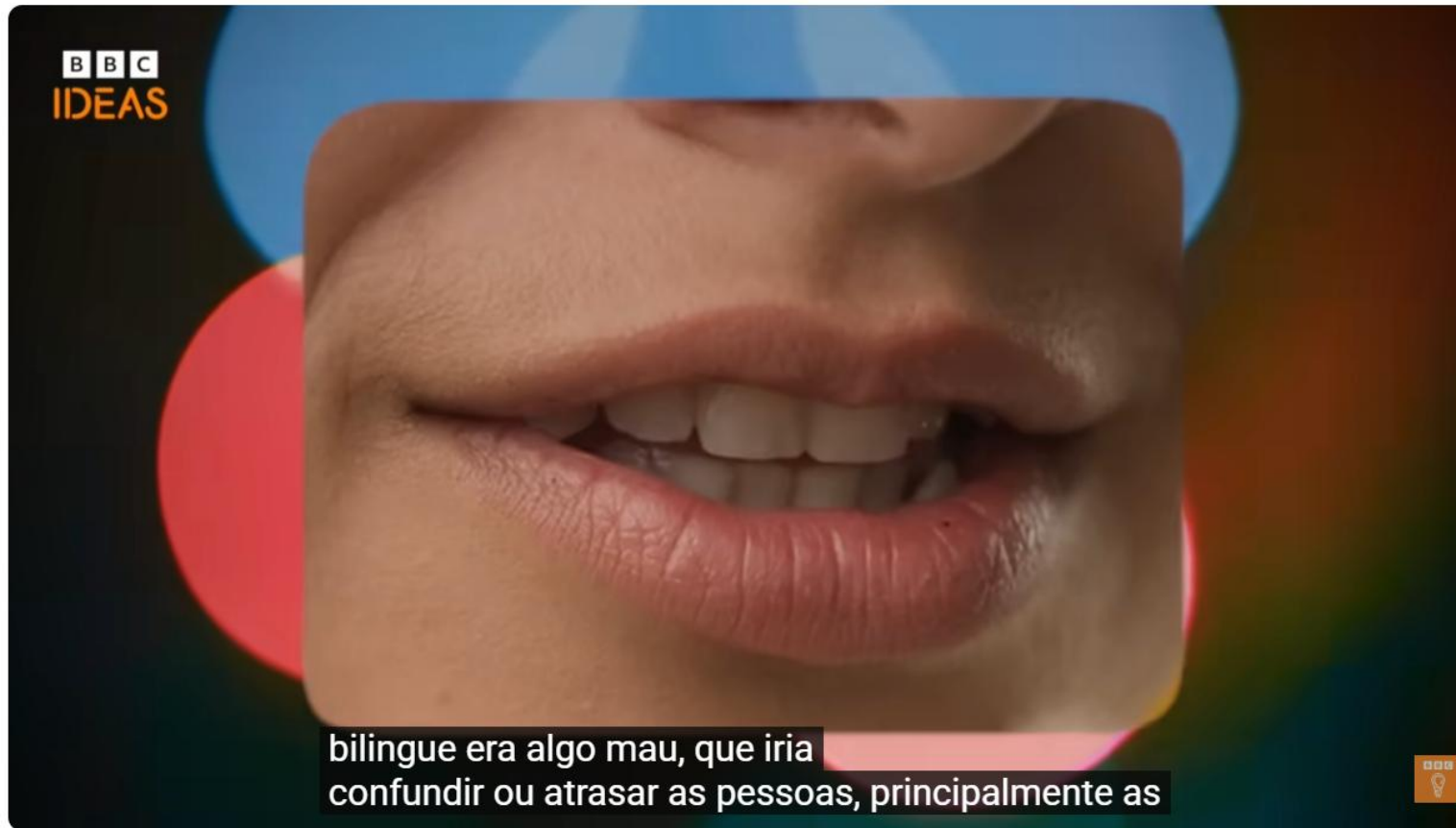
Participants are able to

<https://lhq-blclab.org/>





FALAR DUAS LÍNGUAS TORNA-NOS MAIS INTELIGENTES?



<https://www.youtube.com/watch?v=nzHY-muy2Mw>

FALAR DUAS LÍNGUAS TORNA-NOS MAIS INTELIGENTES?

Los Angeles Times

Bilingualism good for the brain, researchers say

The skill helps improve multitasking and prioritizing, and helps ward off early symptoms of Alzheimer's disease, experts say.

Mail & Guardianonline

On the tip of the tongue

ALOK JHA Mar 18 2011 10:37

0 comments | Post your comment

"Being bilingual has certain cognitive benefits and boosts the performance of the brain, especially one of the most important areas known as the executive control system," said York University psychology professor Ellen Bialystok at the annual meeting of the American Association for the Advancement of Science being held in Washington, DC.

Bilingualism Is Like A Mental Gymnasium For The Brain



FALAR DUAS LÍNGUAS TORNA-NOS MAIS INTELIGENTES?

Bilingualism, Mind, and Brain Lab

Welcome Research Lab Members Selected Publications Selected Talks Recent Media Research Opportunities
Events ▾

Judith F. Kroll



Distinguished Professor

Co-Founder, [Women in Cognitive Science](#)

PI, [NSF PIRE: Translating Cognitive and Brain Science in the Laboratory and Field to Language Learning Environments](#)

<https://bilingualismmindbrain.com/lab-members/judith-f-kroll/>

What is the consequence of parallel activity and competition across the bilingual's two languages?



Juggling may tune brain networks that enable control and build cognitive reserve

BRAIN BENEFITS OF BILINGUALISM



PREVENTING DEMENTIA

Bilingual adults with Alzheimer's take twice as long to develop symptoms as their monolingual counterparts. The mean age for the first signs of dementia in monolingual is 75.5



FOCUSING ON TASKS

Bilingual people display increased concentration on their assignments over their monolingual counterparts. They are more skillful at filtering out irrelevant information



SWITCHING BETWEEN TASKS

Bilinguals are skilled at switching between two assignments, rules and structure — this makes

IMPROVED COGNITIVE SKILLS

Bilingual people have sharper cognitive skills and keep their brain alert and active

DENSER GREY MATTER

Grey matter is responsible for processing language, storing memory and dictating attention span. Bilingual individuals have denser grey matter compared to their monolingual counterparts.



IMPROVED MEMORY

Learning a foreign language involves memorizing rules and vocabulary. This mental exercise improves overall memory, making bilinguals better at remembering lists and sequences



IMPROVED DECISION MAKING SKILLS

Bilingual speakers tend to make more reasonable decisions. Bilinguals are more certain of their choices after thinking in

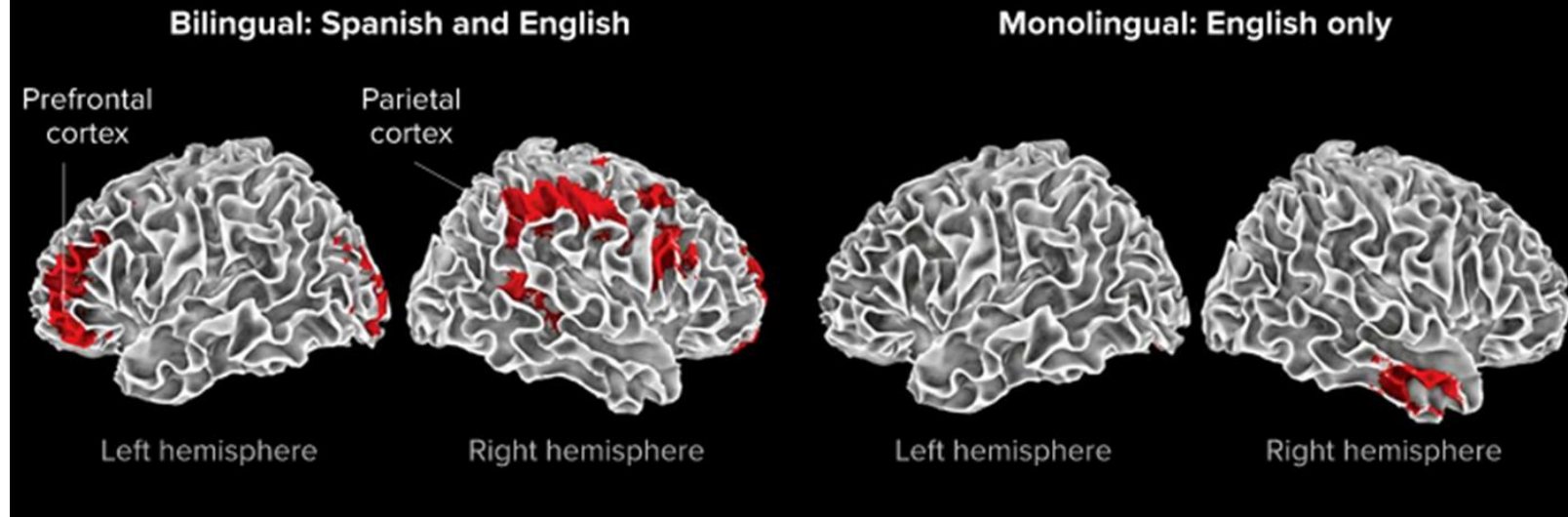


MORE AWARENESS OF LANGUAGE

Learning a second language focuses on grammar and sentence structure. Bilinguals are more receptive to their choices after thinking in their second language.

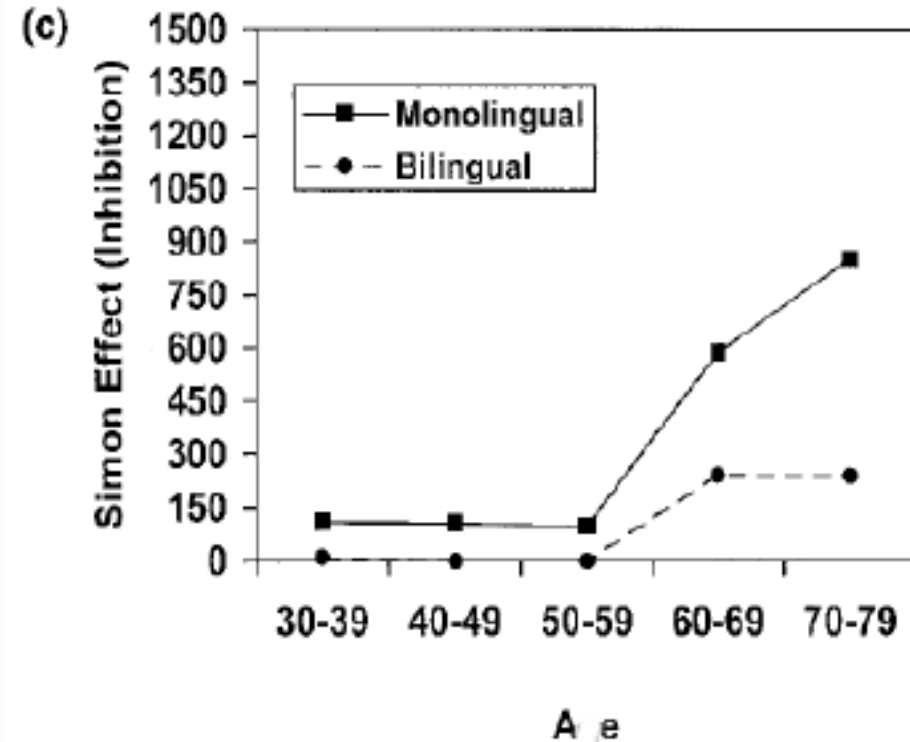
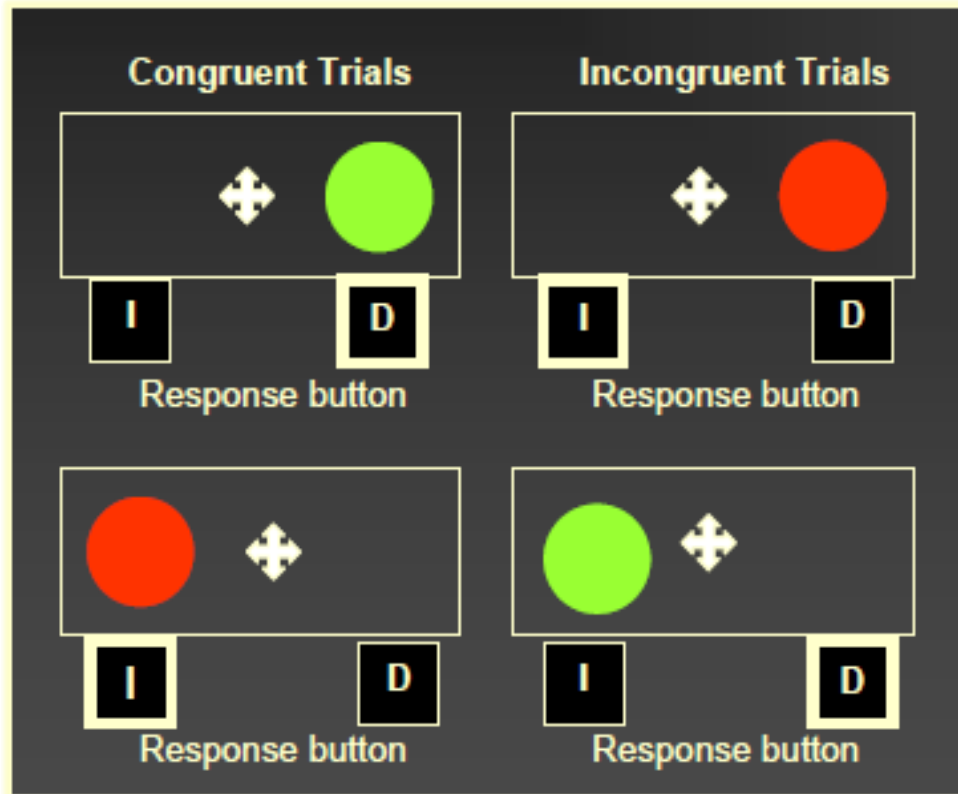
SOURCE: <http://bit.ly/TbXHSZ>

Gray-matter volume in bilinguals vs. monolinguals



- indivíduos expostos a uma L2 no início da vida **mostram maior densidade de tecido cerebral em regiões associadas à linguagem, memória e atenção** (e.g., Della Rosa et al., 2013; Kovelman et al., 2008; Michelli et al., 2004; Olulade et al., 2016; Stein et al., 2017),
- **maior integridade da matéria branca** (e.g., Anderson et al., 2018) e
- maior **conectividade funcional entre regiões de matéria cinzenta** (e.g., DeLuca et al., 2019; Hosoda et al., 2013),
- o que pode explicar as **vantagens que os bilingues demonstram numa vasta gama de tarefas cognitivas** e também o papel protetor que a aprendizagem de mais do que uma língua parece trazer ao funcionamento cognitivo em geral (Bialystok, 2017; Grundy et al., 2017).

EFEITO SIMON BILINGUES



EXISTE MESMO VANTAGEM BILINGUE?

- Meta-análises mostram resultados mistos

- Não há vantagem “universal”
- Efeitos dependem de: idade, tipo de tarefa, contexto, proficiência, NSE, imersão, ordem de aquisição
- Não basta falar duas línguas, o contexto em que o fazemos interessa!
- Crianças bilingues que usam as duas línguas diariamente, em contextos socialmente exigentes, parecem beneficiar mais do que aquelas que têm contacto mais passivo.

Review

> Psychol Bull. 2018 Apr;144(4):394-425. doi: 10.1037/bul0000142. Epub 2018 Mar 1.

FULL TEXT LINKS

Is b
fun



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ISSN: 0033-2909

Psychological Bulletin

2020, Vol. 146, No. 12, 1059–1083
http://dx.doi.org/10.1037/bul0000301

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PubMe

Is Bilingualism Related to a Cognitive Advantage in Children? A Systematic Review and Meta-Analysis

Hilde Lowell Gunnerud, Dieuwer ten Braak,
and Elin Kirsti Lie Reikerås
University of Stavanger

Enrica Donolato and Monica Melby-Lervåg
University of Oslo

Bilingual people are often claimed to have an advantage over monolingual people in cognitive processing owing to their ability to learn and use two languages. This advantage is considered to be related to executive function (EF). However, no consensus exists as to whether this advantage is present in the population or under which conditions it prevails. The present meta-analysis examines the bilingual advantage in EF of children aged 18 years and under for different components of inhibition (hot: rewarding stimuli/cold; neutral stimuli), attention, switching, monitoring, working memory, and planning in 143 independent group comparisons comprising 583 EF effect sizes. The bilingual advantage in overall EF was significant, albeit marginal ($g = 0.06$), and there were indications of publication bias. A moderator analysis showed significant group differences on EF in favor of bilinguals for studies of children from middle-class socioeconomic backgrounds and studies from one specific lab. The EF components of cold inhibition, switching, and monitoring expressed significant bilingual advantages, but monitoring and cold inhibition were affected by publication bias. As for switching, this remained significant after controlling for publication bias. Thus, given the small mean effect size and small-study effects, this meta-analysis gives little support for a bilingual advantage on overall EF. Still, also after the moderator analysis, there was a large heterogeneity of true effects and a large amount of unexplained heterogeneity in the effect sizes. Thus, there might be bilingual advantages (or disadvantages) under conditions that this study is not able to identify through the analysis of 12 moderators.

Public Significance Statement

The hypothesis that bilingual children have a cognitive advantage related to their executive function has been accepted as a truth in both society and research. However, the present meta-analysis, which is based on a large number of studies (143 comparisons and 583 effect sizes), gives little compelling support for the position that a bilingual advantage exists for overall executive function. Importantly, there was large heterogeneity in results and studies of middle-class children and studies using switching tasks, and studies from one lab demonstrated larger effects.

O QUE SABEMOS HOJE?

Cognitivos

+ Controlo atencional em certos contextos

+ Flexibilidade cognitiva adaptativa

Não é aumento de QI

Neurais

+ Plasticidade e eficiência neural

+ Conectividade funcional adaptativa

Não “mais cérebro” mas “diferente cérebro”

Sociais & Socioculturais

+ Empatia cultural & flexibilidade

+ Sensibilidade intercultural

+ Competências comunicativas & sociais

MOMENTO PARA RESPONDER E REFLETIR (MR²)

Falar duas línguas torna-nos mais
inteligentes?

- Aprender línguas não nos torna mais inteligentes no sentido de QI, mas expande a forma como o cérebro processa o mundo, reforça o controlo mental e aumenta a nossa abertura cultural
 - benefícios mais consistentes: flexibilidade cognitiva, controlo atencional, plasticidade neural, ToM (empatia)
- Não existe vantagem cognitiva universal → depende do contexto e do uso.
- Bilinguismo ≠ mais inteligência vs. Bilinguismo = mais ferramentas para pensar e viver no mundo.